GROUP WEEKLY REPORTS

Project Title: Why 2024 Flights Ran Late
Team: Team 10 – DATA-VIZ HACK-A-BIT
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WEEK 6 – OUTLIER DETECTION AND ANALYSIS

During the sixth week, our team focused on identifying and analyzing outliers in the flight delay dataset. Since the dataset contained millions of observations, unusual values could significantly influence statistical summaries and affect the interpretation of visualizations.

We explored different statistical methods for detecting outliers, particularly the Interquartile Range (IQR) method and the Z-score method.

The IQR method was used to identify observations that lie outside the lower and upper boundaries calculated using the first quartile, third quartile, and interquartile range. The Z-score method helped us understand how far individual observations were from the mean in terms of standard deviations.

We applied these techniques to numerical variables, especially departure delay and arrival delay. The team examined the detected outliers to understand their characteristics and potential impact on the dataset.

We also discussed the limitations of outlier detection methods. In flight data, unusually large delays may occur because of operational disruptions, weather conditions, technical issues, or other genuine circumstances. Therefore, automatically removing all outliers would not be appropriate.

Instead, we focused on distinguishing genuine extreme observations from possible data errors. We considered the importance of retaining meaningful information while ensuring that statistical analysis remained reliable.

By the end of the week, we had gained practical experience in outlier detection, understood the significance of extreme values in real-world datasets, and documented our observations for the project.